

Module 5

GKE in GCP



Goals: Understand about Google Kubernetes Engine

- Create and config cluster, node pool
- Access to the cluster
- Deploy workload
- Log monitoring and alert
- Migrate node pool
- GKE Security Posture (NEW)

- **Kubernetes** : Kubernetes is an **open source** container orchestration platform that automates many of the manual processes involved in deploying, managing, and scaling containerized applications.
- Kubernetes was originally developed and designed by engineers at Google. Google was one of the early contributors to Linux container technology and has talked publicly about how **everything at Google runs in containers**.

Here are some things to keep in mind when creating a node pool:

- The node pool name must be unique within your project.
- The machine type must be a Google Compute Engine machine type.
- The number of nodes must be a positive integer.
- The Kubernetes version must be a supported Kubernetes version.
- The Kubernetes network must be a Kubernetes network in your project.
- The Kubernetes subnetwork must be a Kubernetes subnetwork in the Kubernetes network that you selected.

Here are the methods that you can use to connect to a GKE cluster:

- Command-line: You can use the gcloud command-line tool to connect to a GKE cluster. To do this, run the following command:
`gcloud container clusters get-credentials <cluster>`
- Web console: You can use the Cloud Platform Console to connect to a GKE cluster. To do this, click the Connect button in the cluster overview page.
- Kubernetes CLI: You can use the Kubernetes CLI to test the connection to a GKE cluster. To do this, run the following command:
`kubectl get nodes`

Some things to keep in mind when monitoring logs and alerts in GKE:

- The logs for the cluster are stored in Cloud Logging.
- You can use Cloud Logging to view, search, and analyse the logs for the cluster.
- You can use Cloud Monitoring to create alerts for the cluster.
- Cloud Monitoring will notify you when an alert is triggered.

Additional things you can do to monitor logs and alerts in GKE:

- You can use the Cloud Logging API to programmatically access the logs for the cluster.
- You can use the Cloud Monitoring API to programmatically create alerts for the cluster.
- You can use the Cloud Monitoring UI to create dashboards for the cluster.

- The image you sent is a screenshot of the GKE security posture dashboard. This dashboard provides a high-level overview of the security posture of your GKE clusters. It includes information about the following:
 - Clusters: The number of clusters in your project, the number of nodes in each cluster, and the overall health of each cluster.
 - Workloads: The number of workloads running in your clusters, the number of containers in each workload, and the overall health of each workload.
 - Security alerts: Any security alerts that have been raised for your clusters or workloads.

Thank you

